

Back when Bill Gates was Chief Software Architect under Microsoft, he called on employees to make security the primary focus of present and future Microsoft products. The announcement was made in the atmosphere of distrust towards the organisation's products, as every day seemingly brought new exploits and malware to the Microsoft ecosystem. Windows XP SP2 was the first step to rectifying this problem. Microsoft also made a series of changes to its internal structure and to the manner in which the company developed software. A security-focused engineering process called Security Development Lifecycle (SDL) was established within Microsoft to move security into the very essence of software design; all widely deployed Microsoft products must follow these guidelines.

Windows Vista is the first large project to be launched under the aegis of SDL. It includes new or upgraded inbuilt security technologies that actively work to detect and prevent security threats. These changes include the new User Account Control feature, volume encryption using BitLocker, code integrity verification, Windows Defender, Windows Firewall, and a new Security panel. Let's take a closer look at each of these elements and understand how they protect the OS, and what changes they bring.

## 6.1 User Account Control (UAC)

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There are generally two types of users in any operating system—those who can make changes to the system's files and settings, and those who cannot. For Windows XP, the OS would by default grant a user all the rights and privileges required to make changes to the operating system. While this made it easier for a user to install programs, move folders, remove programs, change time zones and so on, it also opened up a huge security hole for malicious programs. Any program that ran in such a high-privileged atmosphere inherited all the rights from the user and was thus just as capable of making system-wide and potentially harmful changes.